Evidence budgeting for edge AI under degraded connectivity

A validation study on synthetic and real disaster data: what adaptivity buys, and where it does not

A fixed rule matches the exact optimizer

4.2 vs 4.4

errors out of 720, at half the bytes

Admissibility collapses under forged metadata

7.0 → 20.2

errors, reaching the unfiltered baseline

A measured link is not a constant one

×8.7

tail latency, at matched mean capacity

The optimizer needs centralisation

+20.0 vs +3.9

errors when coordination is removed

On real data the verdict reverses

+17.2

errors, optimizer over fixed rule, 2 of 3 conditions

The formulation earns its place by naming the quantities and supplying the threshold.
Not the solver: it never beat a greedy alternative on the same utility, on either workload.

Nine synthetic conditions · a real CrisisMMD workload held out by event · five seeds · two pre-registered tests · artifact reproduces byte for byte